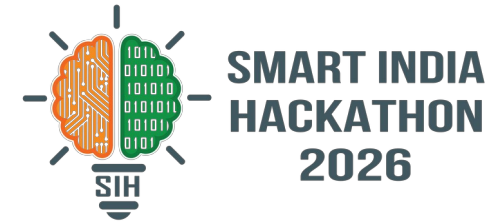


SMART INDIA HACKATHON 2026



ANVESHA

Finds a mobile FSOC beacon in 1.3 s (median, simulated) by deciding **where** and **how wide** to look — then tracks it

- Problem Statement ID : [SIH26169](#)
- Problem Statement Title : [Development of an AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile Free Space Optical Communication \(FSOC\) Terminals](#)
- Theme : [Smart Automation / Space Technology](#)
- PS Category : [Software](#)
- Team ID : _____
- Team Name (Registered on portal) : [Regnum Carya](#)

1.3 s

median acquisition
spiral baselines 6.4–9.2 s

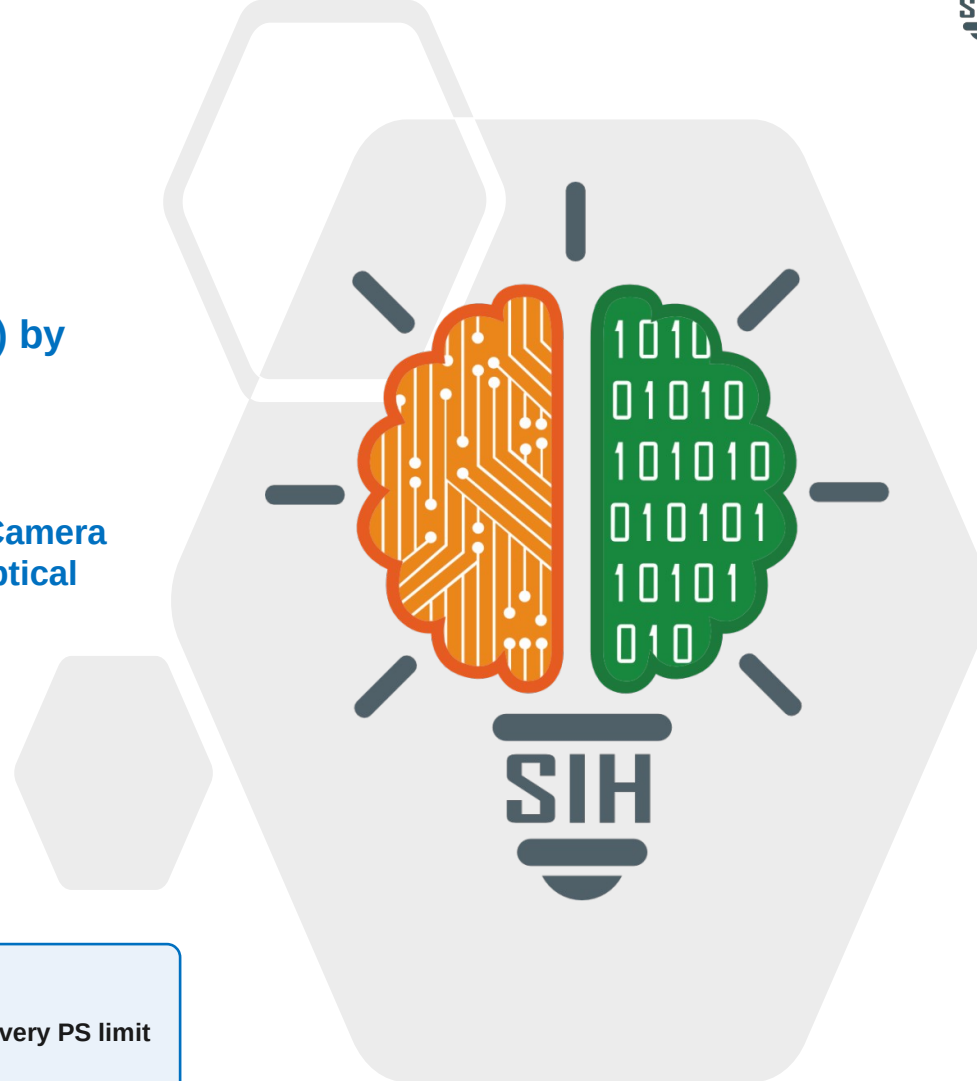
87 %

of runs acquired ≤ 2 s
baselines 12–15 %

11/16

scenarios meet every PS limit
baselines 0/16

Measured in simulation (software-in-the-loop): 16 scenarios $\times$ 3 seeds, identical seeds for all pipelines



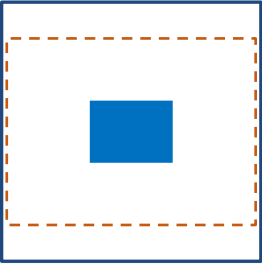
IDEA TITLE

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

ANVESHA: a probability map of the beacon decides **where to point** and **how wide to look**.

Measured (SIL, 16 scenarios × 3 seeds): 87 % of runs acquired ≤ 2 s vs 12–15 % for spiral baselines (slide 4)

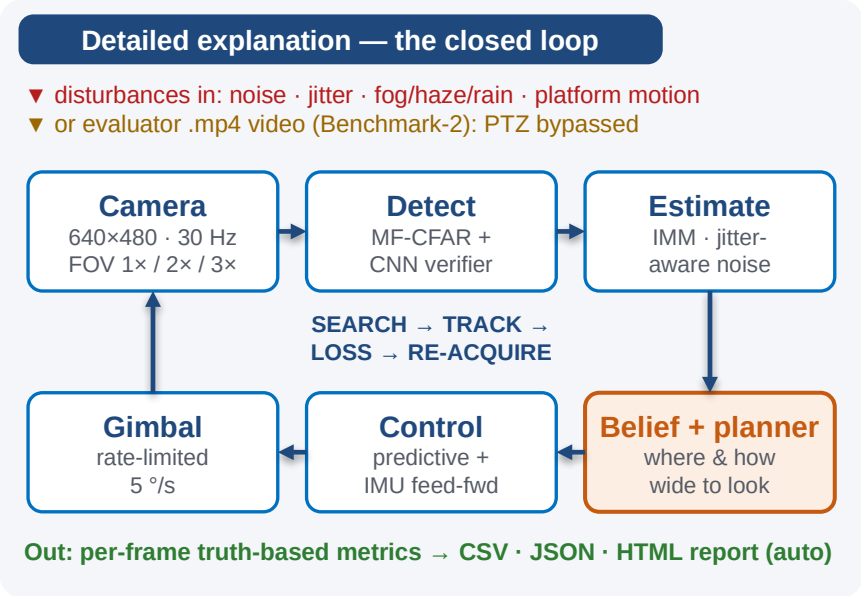
How it addresses the problem



4°×3° FOV = **7.7 %** of the screen
5 °/s gimbal ⇒ ≤ 10° slew in 2 s
Blind raster needs ≈ 40–50° of slew ⇒ **~8–10 s**
Measured spiral baselines A / B: median **7.8 / 9.2 s**
PS asks ≤ **2 s** from a random start

To scale: screen 12.5°×12.5° · ■ 4°×3° base FOV · ▭ 12°×9° wide FOV

Detecting a bright spot is easy. Knowing **where to look** is the real problem.



Innovation and uniqueness

1

Belief-driven search
every empty frame is evidence: probability is removed where the beacon was not seen; next look = best P(detect) per second

2

Disturbance-aware FOV
FOV (1×/2×/3×) chosen by predicted detection probability from the measured noise: go wide only when the beacon stays detectable

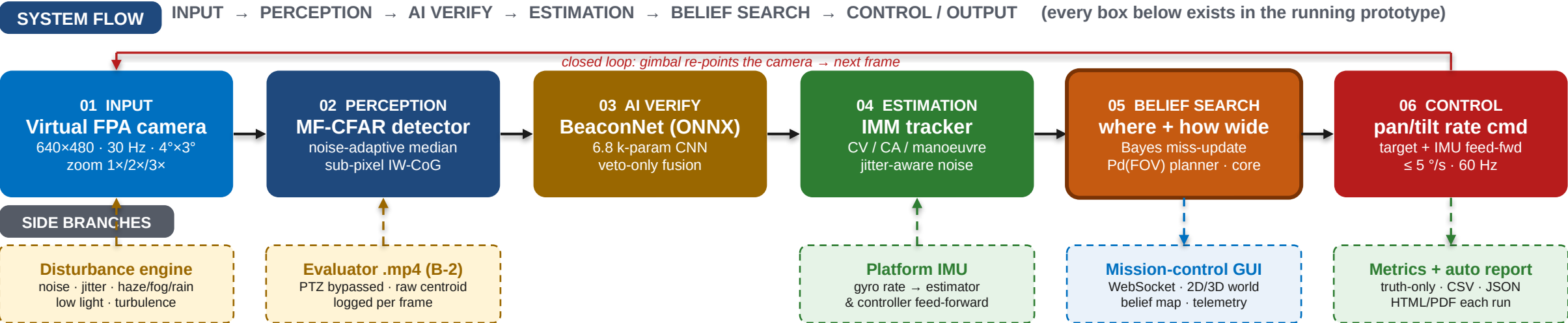
3

One belief, whole lifecycle
on loss the search restarts from the tracker's prediction; jitter-aware estimator stops the gimbal chasing jitter

Typical public approach	Limitation	ANVESHA approach
Fixed spiral / raster at base FOV	Open loop; ignores empty frames; ~8–10 s worst case	Recursive Bayesian belief + next-best-look planner
Wide-FOV mode switched by a fixed rule	SNR loss of a wider FOV is not modelled	FOV picked from a measured-noise detection model
Re-acquire = restart spiral / Kalman coast	Throws away what the tracker knew	Belief re-seeded from IMM predicted mean & covariance

Based on our review of 134 papers and 20 public SIH26169 repositories, we did not identify a coarse-PAT system that combines these three elements.

❖ Methodology and process for implementation (Flow Charts/Images/ working prototype)



❖ Analysis of the feasibility of the idea

Measured in simulation (software-in-the-loop), 3 seeds · 16 scenarios · same seeds for all pipelines · baseline PID gains grid-tuned · laptop CPU only (no GPU used)

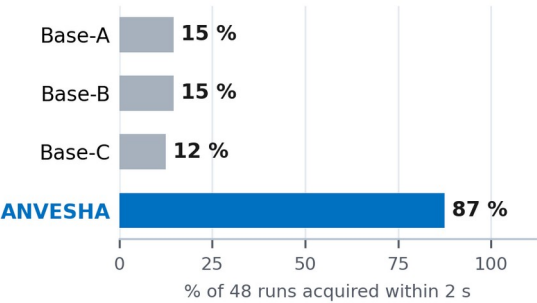
ANVESHA · scenario	Acq. ≤2 s	LOS err	Image err	Loss*	Re-acq	Loop FPS
A Clean sky	3/3 · 1.3 s	1.5	1.5	0 %	—	167
B Gaussian σ=20	3/3 · 1.2 s	1.9	1.9	0 %	—	166
C Poisson	3/3 · 1.1 s	0.8	0.8	0 %	—	55
D Salt & pepper 10 %	3/3 · 1.1 s	6.7	6.7	0 %	—	142
E Jitter ±10 px	3/3 · 1.3 s	4.6	8.9	0 %	—	170
F Platform linear	3/3 · 1.1 s	4.4	4.4	0 %	—	167
F2 Vibration 7/13 Hz	2/3 · 1.4 s	20.5	20.5	0 %	—	168
G Fog	3/3 · 1.3 s	1.5	1.5	0 %	—	136
H Haze	3/3 · 1.2 s	3.5	3.5	0 %	—	136
I Low light	3/3 · 1.2 s	2.2	2.2	0 %	—	54
J Fast manoeuvre	3/3 · 1.3 s	18.6	18.6	0 %	—	168
K Occlusion	3/3 · 1.3 s	16.2	16.2	1 %	0.10 s	147
L Rain	3/3 · 1.4 s	1.9	1.9	0 %	—	124
M Combined	3/3 · 1.2 s	4.9	7.8	0 %	—	122
N All PS maxima	0/3	21.0	23.6	99 %	n/a (1X)	51
O Look-alike distractors	1/3 · 2.9 s	1.5	1.5	0 %	—	170

All PS checks met: ANVESHA 11/16 scenarios · baselines A/B/C 0/0/0
 Limits: acq ≤ 2 s · error ≤ 10 px · loss < 5 % · re-acq ≤ 1 s · ≥ 20 FPS. *excl. scripted occlusion · LOS = excl. camera jitter

Measured gaps — shown, not hidden

- All PS maxima at once (N): acquired in 33 % of runs; look-alikes (O) acquired in 2.9 s (> 2 s) — rejecting static look-alikes costs time.
- LOS error 16–21 px (> 10) under vibration, fast manoeuvre, occlusion.
- Jitter floor: ±20 px white jitter alone = 16.3 px RMS (analytic) — no coarse gimbal can meet 10 px image error. At ±10 px (E): image 8.9 vs LOS 4.6 px.

Same seeds, all pipelines



Ablation — remove one part (10 scenarios × 3 seeds):

- belief → spiral search: runs ≤ 2 s 83 % → 7 %
- adaptive FOV off: runs ≤ 2 s 83 % → 20 %
- jitter-aware R off (E): LOS 4.6 → 19.3 px
- IMU feed-fwd off (F): LOS 4.4 → 5.8 px
- CNN verifier off: false-lock frames 614 → 667 (sum)

Re-acquisition (K, 2 occlusions): mean 0.10 s, max 0.27 s, 0 of 6 unrecovered

Benchmark-2 video rehearsal: centroid RMSE 0.13 px over 300 frames of 2000×2000 @30 fps; 88 FPS incl. decoding

Median acquisition when acquired: ANVESHA 1.3 s · baselines 6.4–9.2 s

Potential challenges and risks	Strategies for overcoming these challenges
Sim-to-real gap	video mode already accepts real footage; tracker needs only frames, encoder / IMU and rate commands → HIL
±20 px jitter floor	report LOS vs image error; jitter-aware R; fine stage (FSM) hand-off
Faint beacon, clutter, look-alikes	joint position × detectability belief; CNN veto + shape tests; static look-alike rejection; beacon modulation ID (next)
CPU budget	ROI processing; CNN only on candidate patches; FPS logged each frame
CNN domain shift	CNN only verifies; classical MF-CFAR stays primary

Implementation status

- Implemented

sim, 16 scenarios, 4 pipelines, belief search, IMM, GUI, video mode, reports
- Prototype

CNN verifier: recall 90 %, FPR 23 % on held-out synthetic patches
- Planned

.exe build · HIL rig · real footage

❖ Potential impact on the target audience

- 1

ISRO / DoS PAT teams: design & regression-test coarse PAT for mobile ground terminals, UAV/HAP relays and LEO links before hardware exists
- 2

Academia & students: seeded, repeatable PAT testbed — no optics lab, camera or gimbal needed
- 3

Indian FSOC start-ups: tracker talks only frames, encoder/IMU and rate commands → swap simulator for a real camera + pan-tilt (HIL)

❖ Benefits of the solution (social, economic, environmental, etc.)

- **Operational** — link set-up: median acquisition 1.3 s vs 6.4–9.2 s for spiral search (simulation) — less dead time per mobile hand-over
- **Economic** — tune and regression-test PAT logic before buying cameras, pan-tilt units or an optical bench
- **Technical** — fog, rain, jitter and vibration sweeps that cannot be repeated outdoors — every run = scenario + seed, truth-logged
- **Strategic / social** — indigenous, auditable coarse-PAT software; open testbed for Indian students and FSOC start-ups
- **Environmental** — fewer vehicle / field trials needed for early tuning



What makes ANVESHA different

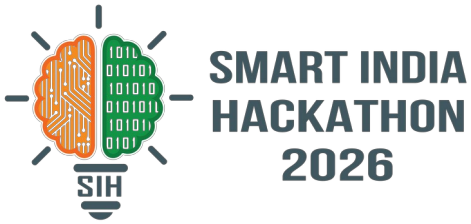
Capability	Typical public approach*	ANVESHA
Probability-map (belief) search instead of a fixed spiral / raster	✗	✓
FOV chosen from measured noise & predicted detectability	✗	✓
Re-acquisition seeded from the tracker's prediction	✗	✓
Online jitter estimate — gimbal does not chase jitter	✗	✓
Platform IMU feed-forward	some	✓
Baselines + ablation on identical seeds	some	✓
Benchmark-2 video (.mp4) bypass mode	✓	✓
GUI + automatic performance report	✓	✓

**majority pattern in the READMEs of 20 public SIH26169 repositories · no team named*

Why acquisition matters: an on-orbit optical link reported a 22 s acquisition time [Wang et al., Opt. Lett. 2023].



RESEARCH AND REFERENCES



❖ Details / Links of the reference and research work

134

papers in a structured literature database (DOIs)

20

public SIH26169 repositories analysed for differentiation

16×4×3

scenarios × pipelines × seeds + 7-way ablation

[1] Kaymak Y. et al., “A Survey on Acquisition, Tracking, and Pointing Mechanisms for Mobile FSO Communications,” IEEE COMST, 2018. doi.org/10.1109/comst.2018.2804323

[2] Abdelfatah R. et al., “A review on PAT approaches in UAV-based FSO communication systems,” Opt. Quantum Electron., 2022. doi.org/10.1007/s11082-022-03968-2

[3] Wang X. et al., “On-orbit space optical communication demonstration with a 22 s acquisition time,” Opt. Lett., 2023. doi.org/10.1364/ol.505966

[4] Friederichs L. et al., “Vibration Influence on Hit Probability During Beaconless Spatial Acquisition,” J. Lightwave Technol., 2016. doi.org/10.1109/jlt.2016.2542918

[5] Wang T. et al., “Probability Descent-Based Equivalent Spot Scanning for Satellite Laser Acquisition...,” IEEE TAES, 2025. doi.org/10.1109/taes.2025.3596959

[6] Skryja P., Barcik P., “Optimal Scanning Pattern for Initial Free-Space Optical-Link Alignment,” Photonics, 2024. doi.org/10.3390/photonics11060540

[7] Wang Q. et al., “Approach for recognizing and tracking beacon in inter-satellite optical communication...,” Opt. Express, 2018. doi.org/10.1364/oe.26.028080

[8] Lim H. et al., “Centroid Error Analysis of Beacon Tracking under Atmospheric Turbulence...,” Remote Sensing, 2021. doi.org/10.3390/rs13101931

[9] Li L. et al., “Noise adaptive fading Kalman filter for free-space laser communication beacon tracking,” Appl. Opt., 2016. doi.org/10.1364/ao.55.008486

[10] Youn W., Myung H., “Robust Interacting Multiple Model With Modeling Uncertainties for Maneuvering Target Tracking,” IEEE Access, 2019. doi.org/10.1109/access.2019.2915506

[11] Bian Q. et al., “Accelerometer-Assisted Disturbance Feedforward Control of an Inertially Stabilized Platform...,” IEEE Sensors J., 2023. doi.org/10.1109/jsen.2023.3259446

[12] Kiesbye J. et al., “Hardware-In-The-Loop and Software-In-The-Loop Testing of the MOVE-II CubeSat,” Aerospace, 2019. doi.org/10.3390/aerospace6120130

SCALABILITY & DEPLOYMENT

- **Hardware-agnostic:** tracker consumes only frames, encoder angles, IMU rates → a camera + pan-tilt adapter replaces the simulator (planned)
- **Terminal-agnostic:** resolution, FOV, zoom set, gimbal limits, noise & atmosphere are YAML parameters
- **Compute headroom:** laptop CPU, no GPU — ≥ 51 FPS closed loop; 2000×2000 video at 88 FPS
- **Regression at scale:** 192-run matrix in parallel across CPU cores; unit tests + fixed seeds

THE ONE THING TO REMEMBER

Detecting a bright spot is easy — knowing where and how wide to look is the problem.

ANVESHA: 1.3 s median acquisition vs 6.4–9.2 s blind search; all PS limits met in 11/16 scenarios (baselines 0/16).

Measured in simulation · failures shown on slide 4 · Based on our review of 134 papers and 20 public SIH26169 repositories, we did not identify a coarse-PAT system that combines belief search, noise-aware FOV and belief-seeded re-acquisition.

PROJECT RESOURCES

QR
add link

Source code

QR
add link

Demo video

Offline demo
run_gui.bat
Benchmark
run_benchmark.bat
B-2 video
run_video.bat

Every number on slides 1–6 is regenerated from results/*.json by the build script.